Analysis & Design of System controls Exam 1

abc 123: eaq 108

q = 1+0+1 = 2

b = 0+8+2 = 10

c = 1+0+8+3 = 12

① $R(s) = \frac{12}{s} - \frac{3(10)}{s+2}$

$\downarrow \mathcal{L}^{-1}$ $\downarrow \mathcal{L}^{-1}$

12 $30\, e^{-2t}$

Ans $\boxed{r(t) = 12 - 30\, e^{-2t}}$